

Day 2

The core components of Linux (Kernel, userspace, init / systemd)

How processes are created and managed

What systemd does and why it matters.

1) Core components of Linux -

Kernel

- It is the heart of Linux.
- It acts as a bridge between hardware and software
- It manages -
 - cpu, memory, processes,
 - devices, File systems, Networking.

User space

- Where user interact & appl's run
- It contains -
 - shell (bash)
 - commands
 - Appl's
 - Libraries
- Appl's in user space cannot directly access hardware
- They communicate with the kernel using system calls.

Init / systemd →

After the linux kernel boots, it starts the first process :

PID 1

Modern linux system use - systemd

Older linux system used - SysVinit

Systemd is responsible for -

- 1) Starting services during boot
- 2) Managing background services
- 3) Restarting failed services
- 4) Managing logs.

2) How processes are created and managed ?

A process is a running program in Linux.

Process creation in Linux.

Linux mainly uses two system calls :

1. fork ()
2. exec ()

Step 1) Parent process -

A parent process starts first - bash

2) fork ()

The parent creates a child process using
fork ()

The child process gets:

- 1) New PID (process ID)
- 2) copy of parent process.

3) `exec()`

The child process loads a new program using `exec()`

`bash` $\rightarrow$ `fork()` $\rightarrow$ `exec()`

Process Management -

Linux kernel manages -

- i) CPU scheduling
- ii) Memory allocation
- iii) Process priorities
- iv) process communication

Each process gets :

- 1) PID (process ID)
- 2) Memory space
- 3) CPU time

3) What systemd does and why it matters?

Systemd is the system and service manager in modern linux operating systems.

It is the first process started after the linux kernel boots.

It runs as

PID 1

Systemd is responsible for managing :

- 1) services
- 2) background processes
- 3) system startup
- 4) logging